Only Waves Can Interfere, Diffract, And Refract

SYNOPSIS

One of the waves' defining characteristics is that they interfere. Wave interference results from the simple addition of waves' amplitudes. That is it. When waves overlap and add, the resultant wave's amplitude reflects the two waves' phase relationship: larger when they are in-phase, smaller when they are out-of-phase. Two waves of identical amplitude and wavelength give twice the amplitude while completely in-phase, and zero — complete cancellation — while completely out-of-phase.

A phase repeats every complete cycle, i.e. the wavelength. So waves are in-phase if they are shifted by exactly one wavelength. If one wave's phase is shifted by half a wavelength with respect to the other, the two are **180° out-of-phase**: the peaks in one align with the troughs in the other, and interference causes complete cancellation.

Recalling two waves rippling away from the sources in the previous PhET simulations: when two waves arrive at a position and overlap, the factor that decides constructive or destructive is **Δd**, the difference in distances from the sources. Integer multiples of a wavelength → constructive. Multiples of a wavelength plus a half → destructive.

Diffraction is a result of wave interference, with one added detail: the wave goes through an aperture, and diffraction describes how it propagates after it does. Why do we care? Diffraction lets waves reach behind obstacles — it is how you hear sound around a corner, pick up radio over a hill, or get a phone signal in a space full of things.

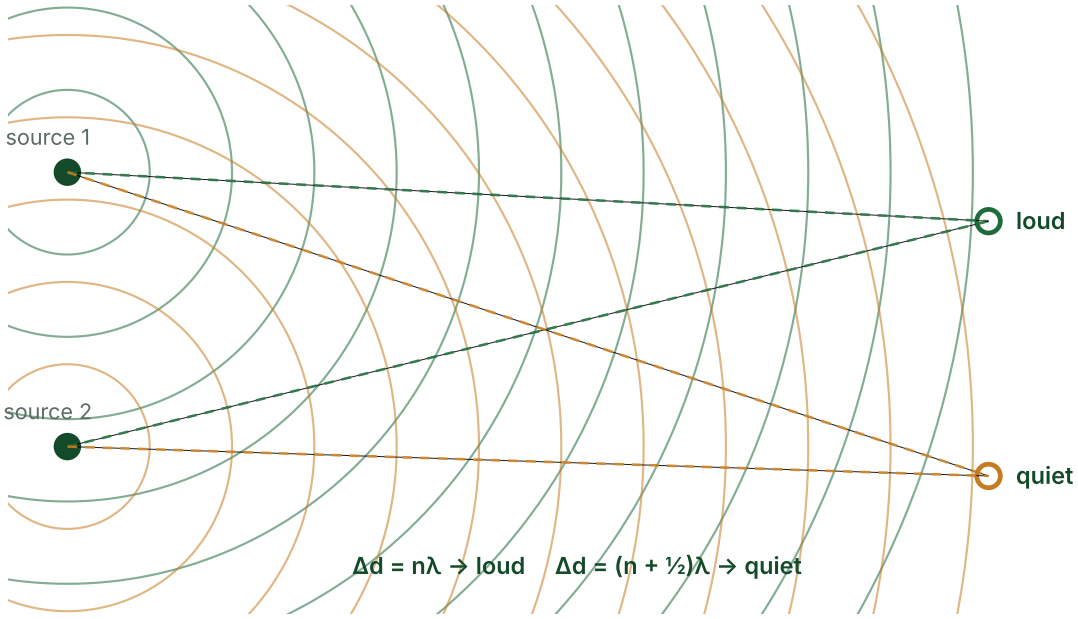


Figure 1. Two sources in step. At any point the only thing that decides loud or quiet is **Δd**, the difference between the two path lengths — a whole number of wavelengths and the peaks arrive together, a whole number plus a half and a peak meets a trough. Nothing here mentions water, air or light, which is the point of measuring it in all three.

PERFORMANCE EXPECTATIONS

Describe constructive and destructive wave interference using waves' phase relationships, and connect those phase relationships to the wavelength. Describe diffraction using wave interference, recognizing that waves spread out off the surfaces and edges they sweep over, and that a large aperture with a short wavelength gives the clearest fringe pattern.

Part I — Interference

1. Open the PhET *Wave Interference* sim, Interference panel, Water. Leave the defaults (frequency middle, amplitude max, separation 3 cm). Put the grey and black probes where the traces are completely **in-phase** — black on the first bright ring, grey on the second. Do the traces confirm it?

2. What is the wavelength?

3. What is the frequency? (Find the period first. How?)

4. What is the amplitude? (One vertical tick mark is 1 cm.)

5. What is the speed of the wave?

6. Black probe on the first bright ring, grey on the second. What is the difference in distances from the two water drops?

7. Move the grey probe to the third bright ring. What is the difference now?

8. How are Δd_1 and Δd_2 related to the wavelength?

9. Now place the probes so the traces are completely **out-of-phase** — black on the first bright ring, grey on the first dark ring. Do the traces confirm it?

10. Measure that difference, then move the grey probe to the second dark ring and measure again.

11. How are Δd_3 and Δd_4 related to the wavelength?

12. Show from your in-phase answers that $\Delta d = n\lambda$, n = 0, 1, 2…

13. Show from your out-of-phase answers that $\Delta d = (n + ½)\lambda$.

14 Turn on the second tap. Without pausing, put the black probe anywhere outside the fuzzy area and the grey probe wherever you see fuzziness. How can you say the black probe shows **constructive** interference? And the grey probe **destructive**?

15 **Water.** Measure top-tap and bottom-tap distances to each probe and take the differences.

16 Show that constructive happens where $\Delta d = n\lambda$, and destructive where $\Delta d = (n + ½)\lambda$. Explain using the phase relationship.

17 **Sound.** Switch to the sound wave and find the wavelength, then repeat the measurement.

18 **Light.** Switch to the light wave and do it once more.

19 Three media — water, air, vacuum. Transverse, longitudinal, electromagnetic. What survived unchanged across all three?

Your own answer — nothing to check against.

Part II — Diffraction

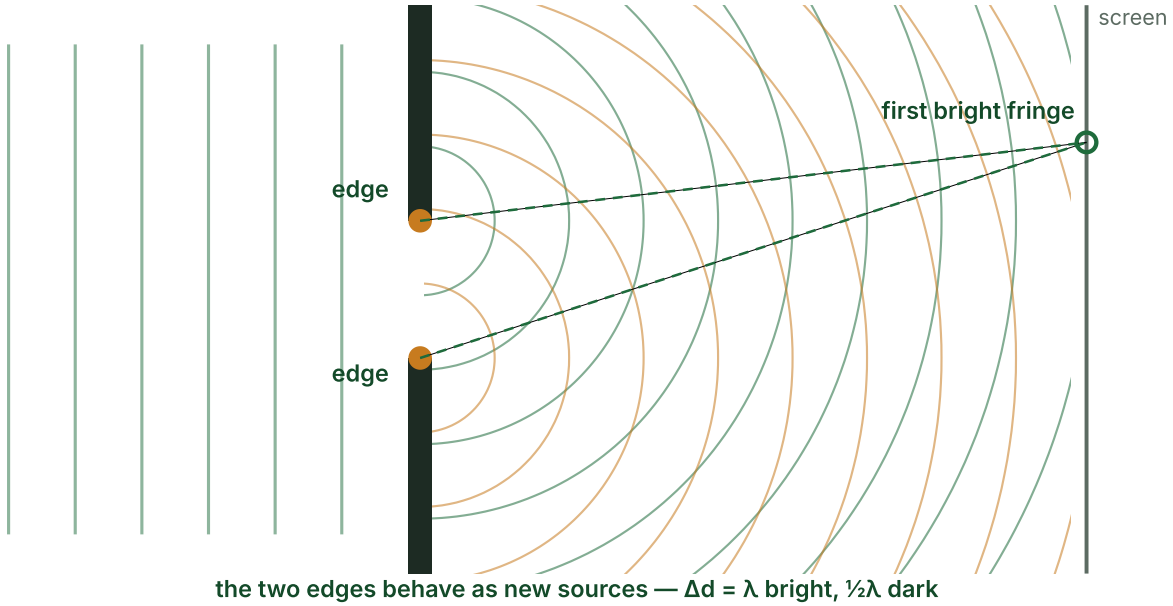


Figure 2. A flat wave meets a slit. Treat the **two edges** as new sources and diffraction becomes the same problem as Figure 1: the path difference from the two edges to a point on the screen is λ at a bright fringe and ½λ at a dark one. That is why a wider slit and a shorter wavelength sharpen the pattern.

1. Run the PhET sim, **Slits** option, Light Generator. Check Screen and Intensity. Slide the frequency all the way left (reddest, longest wave), aperture about 1000 nm from the generator, One Slit. Set the slit width to maximum (1600 nm), send the planar waves, then set it to minimum (200 nm) and send again. Describe the difference in **wavefront shape**.

2. Describe the difference in the **intensity** between the largest and smallest apertures.

3. Repeat with the frequency slid all the way right (blue/purple, shortest wave). Wavefront shape?

4. And the intensity?

5 From the intensity profile, when do you see the **clearest fringes**?

6 The peaks in the intensity profile are **bright fringes**, the valleys **dark fringes**. Explain those terms using constructive and destructive interference.

7 What are the wavelengths of the reddest and bluest frequencies? You need these below.

8 **This statement is critically important:** diffraction is essentially understood by imagining the **two edges of the hole to be new sources of waves**. Set Two Slits, slit width minimum (200 nm), separation 1500 nm, and observe. What are the similarities between the one-slit and two-slit **bright** fringes, and why?

9 What are the differences, and why?

10 Now the **dark** fringes in both profiles. Similarities, and differences?

11 **One Slit**, width back to maximum (1600 nm). Measure from the bottom edge to the first top **bright** fringe, and from the top edge to the same fringe. Take the difference and compare it to the wavelength.

12 Repeat for the first top **dark** fringe.

13 **Two Slits**, width minimum (200 nm), separation 1500 nm. Measure from the top hole and from the bottom hole to the first top **bright** fringe.

14 Repeat for the first top **dark** fringe.

15 Returning to the statement — “the diffraction phenomenon is essentially understood by imagining the two edges of the hole to be new sources of waves” — explain why it is accurate, using your results above.

Your own answer — nothing to check against.

16 Research **secondary wavelets**, coined by the Dutch scientist Huygens. Draw an illustration and describe them in a few sentences.

Your own answer — nothing to check against.

17 Expand on the statements above describing diffraction using the idea of secondary wavelets.

Your own answer — nothing to check against.

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